

Week 7 Board: Software Code Work

Board Content

	To Do	In Progress	Done
Core work	Han.Zhao will review the whole software workflow and check whether the front-end pages match the backend model process. Siyuan.Chen will continue fixing U-Net model loading and mask output issues. Haoran.Niu will check segmentation and clustering parameter logic. Jianchen.Zheng will work on the connection between feature extraction and thickness prediction. Guang.Yang will verify LAB color, centroid, and related feature calculation. Guobao.Yao will improve preprocessing for invalid labels and abnormal image values. Rongzhen.Gu will test the updated code on WS2 and MoS2 sample images. Yixiang.Lin will improve the detection UI, especially batch image import and result display.	At this stage, we are mainly focusing on code optimization and the connection between the interface idea and the backend model code. The U-Net training code is being cleaned, including image and mask loading, 512 x 512 resizing, binary mask conversion, IoU evaluation, and best model checkpoint saving. The UI pages are already designed, but the complete upload-to-result connection still needs further work.	The main UI pages have been completed, including the homepage, login and registration pages, material model selection page, material database, and detection workspace. The U-Net training script has also been organized into a clearer function, so it can be called by server.py more easily in the future. We have also confirmed that the current script is mainly for training and evaluation, not the whole web system yet.
Expedite Task	Before the next demonstration, we need to make one simple test path clearer: select a material model, prepare a microscope image, run the backend U-Net code, and return at least a basic segmentation output or performance result. We also need to organize screenshots and code evidence for the development log.	The team is checking which detection settings are already supported by the current code and which ones are still planned. For example, U-Net training and IoU evaluation are already included, while SLIC, Mean Shift, morphology, contour filtering, and final thickness display still need more integration.	We clarified the relationship between the Chinese web interface and the Python training code. The interface is for the domestic university entrepreneurship competition, so it is designed in Chinese. The backend code is the model part behind the detection page. This makes our next development step much clearer.

Short student summary

This week was mainly about software code problems. We did not focus on buying equipment or changing the whole design. We tried to make the model training code more stable and to understand how it should connect with the web pages.

The most important progress is that the team now knows which part is UI, which part is U-Net training code, and which functions still need to be connected later. This makes the next week's work easier to divide.

Individual task notes for this week

Name	This week task
Han.Zhao	Reviewed the overall software workflow and checked how the user interface should connect with the backend model process.
Siyuan.Chen	Worked on U-Net related debugging, especially model loading, output masks, and stable segmentation results.
Haoran.Niu	Checked segmentation and clustering parameter logic, and helped identify which settings are not fully connected yet.
Jianchen.Zheng	Focused on the connection between feature extraction and thickness prediction, including data transfer between modules.
Guang.Yang	Verified feature calculation code, including LAB color values, centroid coordinates, and derived values used for prediction.
Guobao.Yao	Improved preprocessing logic for abnormal values, invalid labels, and inconsistent input formats.
Rongzhen.Gu	Tested the updated code with WS2 and MoS2 sample images, and compared the result after debugging.
Yixiang.Lin	Improved UI-related parts, especially image import, detection workspace layout, and clearer result display.